

THE CHINESE UNIVERSITY OF HONG KONG

Master of Science in FinTech Programme

Faculty of Engineering

2024/2025

Term 1/2

Course Code & Title: FTEC 5910/5920 Financial Technology Project I/II 金融科技論文 I/II

Professor-in-Charge	Prof. CHAN Chun Kwong	
Academic Supervisors	Department of Computer Science and Engineering	Department of Information Engineering
	Prof. LEE Jimmy H.M.	Prof. CHEN He Henry
	Prof. LO Eric	Prof. LAU Wing Cheong
	Prof. KING Kuo Chin Irwin	Prof. YAN Zhenyu
	Prof. LI Yu	
	Prof. PAN Jialin Sinno	Department of Systems Engineering and Engineering Management
	Department of Decisions, Operations and Technology	Prof. CHAN Chun Kwong
	Prof. LI Tianyi Tim	Prof. CHEN Nan
	Prof. LIU Ruixuan	Prof. AHN Dohyun
	Department of Finance	Prof. CHENG Hong
	Prof. CHEW Seen Meng	Prof. NGUYEN Viet Anh
	Prof. JEUNG Jinoug	Prof. WU Xixin
	Prof. JIANG Wenxi Griffin	Prof. YANG Chen
	Prof. KUONG Chi Fong John	Prof. JIA Yanwei
	Prof. LI Gang	Centre for Financial Engineering
	Prof. WHELAN Paul Kevin	Dr. LEUNG Philip
	Prof. XIE Wensi	Prof. HO Kyson
	Department of Mechanical and Automation Engineering	
	Dr. HAN Dongkun	
Office Hours	By appointment	

Industrial Supervisors		
Sponsoring Organizations	FinTech Companies	Banks and Financial Services
	A.I. Phoenix Technology Co., Ltd. (@HKSTP)	Bank of Communications Co., Ltd.
	AlgoOne (@Cyberport)	China Citic Bank International
	Arrakis Green FinTech (@HKSTP)	Habib Bank Zurich (Hong Kong) Limited
	Digital Financial Services Research Center Limited (@Cyberport)	Hang Seng Bank
	GalaEdge (@Cyberport)	OneInfinity by OneDegree
	PaySmart Capital Limited (@Cyberport)	The Hongkong and Shanghai Banking Corporation Limited (HSBC)
	PFCC	
	RPA HK LTD (@Cyberport)	Others
	Spendsy Limited (@HKSTP)	Hong Kong Arts Investment Company Ltd. (@HKSTP)
	Theo Quant Limited (@Cyberport)	Hong Kong International New Economic Research Institute Ltd (@HKSTP)
	TokenSphere (@Cyberport)	Nuance Pharma Limited (@HKSTP)
	Uncle2 Fintech Ltd	SENtoR Education Company Limited (@HKSTP)

Teaching Assistant (FTEC5910)	
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Teaching Assistant (FTEC5920)	
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Office Hours	By appointment

Class hours and venue

Not Applicable. Students will be working on campus, remotely and at sponsoring company's normal work location as required.

Course Description & Content

The courses, Financial Technology Project I & II, involve a significant research and/or development project under the supervision of an academic staff in any area of financial technology. Unless otherwise approved by the Professor-in-Charge, the project should involve an industrial partner with a co-supervisor from the industry. The co-supervisor will be appointed as CEFAR Fellow during the duration of the project.

At the end of the course, the student must give a presentation to the academic and industry supervisors, and submit a report detailing the findings and deliverables. The student's grade will be determined by the presentation and the report.

Unless otherwise approved by the Programme Director MSc FinTech, the credits for Financial Technology Project I are recognized if and only if the student takes also Financial Technology Project II in a subsequent semester.

If a student has prior industrial background and has already secured a project topic with an industrial practitioner and the agreement of an academic staff, Financial Technology Project I can be taken in the first semester of studies. Otherwise, the two project courses can be taken in any consecutive semesters, including the summer term.

Students are encouraged to find industrial projects by themselves and talk to various academic staffs for project ideas before enrolling in the courses.

金融科技論文I & II是一門由教授指導，在任何金融科技領域上進行一項研究或開發的項目。除得到本科教授的批准，該論文研究需要有一個業界的公司合作夥伴，而且有一個業界的共同指導老師。

在完成課程前，學生需要向學術和行業指導老師作口頭報告，並提交一份報告，詳細說明研究結果。學生的總成績由口頭報告和研究結果報告的成績合計而成。

除得到課程主任的批准，所有學生需於其後的學期修讀金融科技論文II，金融科技論文I的學分才會正式獲批承認。

如學生已有相關行業背景或經驗，並與行業從業者確定了研究題目和得到學術指導老師同意，該學生可於首學期修讀金融科技論文I。除此以外，金融科技論文I & II需要於兩個連續的學期內（包括夏季學期）修讀完成。

修讀本科前，鼓勵學生可先行尋找有興趣的研究題目和與不同學術人員探討研究方向。

Learning Outcomes

At the end of the course of studies, a student should be able to

1. Analyze and solve a significant and practical problem in financial technology;
2. Understand and carry out basic project management;
3. Give a fluent technical presentation; and
4. Write a report detailing the findings and deliverables of the project.

Learning Resources

As provided by the Industrial Supervisor and Academic Supervisor
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Learning Activities

Activities	Number of Hours		Nature of Activities
Review Meetings	In class		NA
	Outside class	Regular review meetings with Academic and Industrial Supervisors to resolve problems in the research/development project	M
Project	In class		NA
	Outside class	On campus working and at sponsor company's normal working location as required	M

Assignment Submission Due Dates:

FTEC5910 (Term 1)

	Assignment	Due Date of submission at 23:59 on
Planning submission	A. A report containing the project goal, significance of the project, problem statement, proposed solutions (deliverables), and proposed timelines. B. planning form	27 Sep, 2024 (Friday)
Midterm submission	midterm progress check form	1 Nov, 2024 (Friday)
End of Term submission	final report and final progress check form	6 Dec, 2024 (Fri)
End of Term Presentation	Presentation to academic and industrial supervisors. The presentation should be 30 min. inclusive of 5 min. for Q&A.	Weeks of 9- 20 Dec, 2024

FTEC5920 (Term 2)

	Assignment	Due Date of submission at 23:59 on
Planning submission	planning form	24 Jan, 2025 (Fri)
Midterm submission	midterm progress check form	28 Feb, 2025 (Fri)
End of Term submission	final report and final progress check form	4 April, 2025 (Fri)
End of Term Presentation	Presentation to academic and industrial supervisors. The presentation should be 30 min. inclusive of 5 min. for Q&A.	Weeks of 7-18 Apr, 2025

Assessment Scheme

The courses, FinTech Project I (FTEC5910) & II (FTEC5920), involve a significant research and/or development project under the supervision of an academic staff in any area of financial technology. The project should involve an industrial partner with a co-supervisor (CEFAR Fellow) from the industry.

At the end of the each course (FTEC5910 and FTEC5920), the student must give a presentation to the academic and industry supervisors, and submit a report detailing the findings and deliverables. The student's grade will be determined by the presentation, including a demo, and the report. The grading components, their weight, rubrics and descriptors are provided below.

For Academic Supervisor

Oral presentation (10%)

Mark Range	Description
9-10	Excellent: the slides are well prepared; speaks excellent English and explains clearly; able to answer questions with adequate explanations; the presentation is engaging; manages time very well.
7-8	Good: good overall, but improvement is needed in some aspects.
6	Fair: fair overall and significant improvement is needed in some aspects.
5	Pass: meets the lowest expectation, but significant improvement is needed in many aspects.
0-4	Failure: fails to meet the lowest expectation in many aspects.

Report Writing (10%) (Evaluation is based on the final report)

Mark Range	Description
9-10	Excellent: The report has good structure and is well written; it contains adequate review of the literature, clear presentation of the problem statement and the solution approach, and insightful discussion of results.
7-8	Good: good overall, but improvement is needed in some aspects.
6	Fair: fair overall and significant improvement is needed in some aspects.
5	Pass: meets the lowest expectation, but significant improvement is needed in many aspects.
0-4	Failure: fails to meet the lowest expectation in many aspects.

Contributions (20%)

Mark Range	Description
18-20	Excellent: has done a lot of work and the contributions are novel and significant; the solution is practical.
14-17	Good: has done a lot of work and the contributions are of some significance; the solution is quite practical.
12-13	Fair: has done some work but is more of the nature of routine implementation; the contributions are limited.
10-11	Pass: has not done enough but meets the lowest expectation.
0-9	Failure: has done nothing or very little and cannot meet the lowest expectation.

Progress (14%) (Evaluation is based on the reported progress from submitted reports and forms as well as the observed progress during meetings)

Mark Range	Description
13-14	Excellent: the student is well-motivated; excellent planning; working hard and making continuous progress; progress far exceeds the supervisor's expectation at the final evaluation time.
10-12	Good: the student is motivated; good planning; working hard and making continuous progress; good progress at the final evaluation time.
8-9	Fair: fair progress overall, but improvement is needed.
6-7	Pass: meets the lowest expectation, but significant improvement is needed.
0-5	Failure: fails to meet the lowest expectation.

Timely Submission (6%)

There are three submissions for a semester with 2 points for each submission.

Mark Range	Description
2	Timely submission of all the required documents
1	Fail to submit all the required documents on time, but submit all of them within one day after the deadline.
0	Fail to submit all the required documents on time and within one day after the deadline.

For Industrial Supervisor

Oral presentation (10%)

Mark Range	Description
9-10	Excellent: the slides are well prepared; speaks excellent English and explains clearly; able to answer questions with adequate explanations; the presentation is engaging; manages time very well.
7-8	Good: good overall, but improvement is needed in some aspects.
6	Fair: fair overall and significant improvement is needed in some aspects.
5	Pass: meets the lowest expectation, but significant improvement is needed in many aspects.
0-4	Failure: fails to meet the lowest expectation in many aspects.

Report Writing (10%) (Evaluation is based on the final report)

Mark Range	Description
9-10	Excellent: The report has good structure and is well written; it contains adequate review of the literature, clear presentation of the problem statement and the solution approach, and insightful discussion of results.
7-8	Good: good overall, but improvement is needed in some aspects.
6	Fair: fair overall and significant improvement is needed in some aspects.
5	Pass: meets the lowest expectation, but significant improvement is needed in many aspects.
0-4	Failure: fails to meet the lowest expectation in many aspects.

Contributions (20%)

Mark Range	Description
18-20	Excellent: has done a lot of work and the contributions are novel and significant; the solution is practical.
14-17	Good: has done a lot of work and the contributions are of some significance; the solution is quite practical.

12-13	Fair: has done some work but is more of the nature of routine implementation; the contributions are limited.
10-11	Pass: has not done enough but meets the lowest expectation.
0-9	Failure: has done nothing or very little and cannot meet the lowest expectation.

Requirements for Submissions:

In each semester, there are three required submissions.

Planning submission	For FinTech Project I, the planning submission consists of two parts. Part 1 is a single report file, which should contain the project goal, significance of the project, problem statement, proposed solutions (deliverables), and proposed timelines. In Part 2, the student needs to fill out the planning form, which contains the tasks that he or she is responsible for in the first term plus the expected achievements up to the midterm and the final. For FinTech Project II, only the planning form is required. The student fills out a new planning form, which contains the tasks that he or she is responsible for in the second term plus the expected achievements up to the midterm and the final. The new planning should take into consideration the suggestions and comments from the academic supervisor and the industrial supervisor in FinTech Project I.
Midterm submission	The student should fill out the progress check form, in which the actual achievements for each planned task up to the midterm are specified and the completion percentage is reported. If the progress falls behind the schedule, justification needs to be provided.
Final submission	The final submission consists of two parts. Part 1 is a single report file, which should provide an introduction to the project (background, motivation and goal), survey the existing literature on the topic, state the problems to be addressed, present the solution approach and the results obtained. A summary of contributions and future research plan should also be given. The format of the report should follow published academic papers in engineering or business. In Part 2, each student should fill out the progress check form, in which the actual achievements for each planned task up to the final are specified and the completion percentage is reported. If the progress falls behind the schedule, justification needs to be provided.

Note: Any student who fails to submit the required reports and forms will receive incomplete grading for the project. Late submission will not only lose points on “timely submission”, but also cause loss of points in “progress” as it indicates poor progress.

Grade Descriptors:

A, A-: EXCELLENT – exceptionally good performance far exceed expectation in all or most of the course learning outcomes. Demonstration of superior understanding of the subject matter, ability to analyze problems and apply extensive knowledge and skillful use of concepts and materials to derive proper solutions.

B+, B, B-: GOOD – good performances in all course learning outcomes and exceed expectation in some. Demonstration good understanding of the subject matter, ability to use proper concepts and materials to solve most of the problems encountered.

C+, C, C-: FAIR – adequate performance in all course learning outcomes. Demonstration of adequate understanding of the subject matter, ability to solve simple problems.

D+, D: MARGINAL – performance barely meet the expectation in all or at least the essential course learning outcomes. Demonstration of partial understanding of the subject matter and ability to solve simple problems.

F: FAILURE – performance does not meet expectation in most the course learning outcomes. Demonstration of serious deficiencies and shall retake the course.

Class Policy

1. All course materials and announcements will be posted on Blackboard, which will be setup per academic supervisor (<http://blackboard.cuhk.edu.hk>). There is no separate website for this course. Please check Blackboard and your university email account regularly.
2. All assignments have to be submitted by uploading to Blackboard by the due dates specified.

Academic honesty and plagiarism

Attention is drawn to University policy and regulations on honesty in academic work, and to the disciplinary guidelines and procedures applicable to breaches of such policy and regulations. Details may be found at <http://www.cuhk.edu.hk/policy/academichonesty/>.

With each assignment, students will be required to submit a signed **declaration** that they are aware of these policies, regulations, guidelines and procedures.

- In the case of group projects, all members of the group should be asked to sign the declaration, each of whom is responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic contents.
- For assignments in the form of a computer-generated document that is principally text-based and submitted via VeriGuide, the statement, in the form of a receipt, will be

issued by the system upon students' uploading of the soft copy of the assignment.

- Students are fully aware that their work may be investigated by AI content detection software to determine originality.
- Students are fully aware of the AI approach(es) adopted in the course. In the case where some AI tools are allowed, students have made proper acknowledgment and citations as suggested by the course teacher.

Assignments without a properly signed declaration will not be graded by teachers.

Only the final version of the assignment should be submitted via VeriGuide.

The submission of a piece of work, or a part of a piece of work, for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration to this effect shall be regarded as having committed undeclared multiple submissions. It is common and acceptable to reuse a turn of phrase or a sentence or two from one's own work; but wholesale reuse is problematic. In any case, agreement from the course teacher(s) concerned should be obtained prior to the submission of the piece of work.

The copyright of the teaching materials, including lecture notes, assignments and examination questions, etc., produced by staff members/ teachers of The Chinese University of Hong Kong (CUHK) belongs to CUHK. Students may download the teaching materials produced by the staff members/ teachers from the Learning Management Systems, e.g. Blackboard, adopted by CUHK for their own educational use, but shall not distribute/ share/ copy the materials to a third-party without seeking prior permission from the staff members/ teachers concerned.

Use of generative AI tools

Approach 1 (by default) – Prohibit all use of AI tools

In assessing the level of achievement of learning outcomes and students' performance, students are expected to produce their own work independently without any collaboration with the use of AI tools.

All use of AI tools is prohibited in assignments and assessment tasks

For assignments and assessment tasks that count towards the final course grades, students are not allowed to submit work which is produced with the collaboration of or supported by the use of any generative AI tools (e.g. ChatGPT)*.

Any breach of the regulations will be considered an act of academic dishonesty and will be handled according to the University's *Procedures for Handling Cases of Academic Dishonesty*.

In case of queries, students should seek advice from the course teacher.

It is each student's responsibility to understand and abide by the policy and regulations of the Honor Code of the Chinese University of Hong Kong. Details may be found at <http://www.cuhk.edu.hk/policy/academichonesty/>. Any breaches of such policy and regulations such as plagiarism or cheating may result in serious disciplinary actions for the students involved.